

Orthopolity as a resource-allocation hypothesis

Reference measures, counterexamples, and an exploratory assessment of aquatic size spectra

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Abstract

Orthopolity proposes that equally allocated resources can produce unequal abundances: classes whose objects require more resource contain fewer objects. We reconstruct this idea as a conditional allocation hypothesis and distinguish it from a universal explanation of power laws. The defining accounting identity depends on a reference measure. With resource cost proportional to size raised to power d , equal resource per logarithmic size interval implies a count-density exponent $d + 1$; equal resource per linear interval implies exponent d . Conservation alone selects neither allocation. We examine frozen earthquake and solar-flare catalogues, a published ocean biomass reconstruction, and 1,300 aquatic size-spectrum estimates from 16 studies. Earthquake magnitudes imply a Gutenberg–Richter coefficient of 0.998, far from the 1.5 required by logarithmic equipartition of the chosen energy proxy. Solar-flare estimates also disagree with the joint allocation-and-scaling prediction on the selected domain. Aquatic normalized biomass slopes have median -1.015 , close to the predicted -1 , but their study-block bootstrap interval does not establish the planned equivalence. A range-metadata audit identifies implausible bounds in 377 records. Excluding them, 9.3% of remaining reported slopes meet a factor-1.25 drift criterion; this is not a count of statistically equivalent systems. We show why centred slopes, fitted dispersion, and conservation cannot establish ensemble equipartition. Resource symmetry supplies a possible physical postulate for equal expected allocations, whose empirical standing does not require a deeper mechanism. Orthopolity supports a useful synthesis and testing framework; the available evidence does not establish a new natural or cosmological law.

1. Introduction

An allocation can be equal when counted in resources and unequal when counted in objects. For example, equal biomass in successive body-mass classes requires fewer large organisms. This observation motivates the term *orthopolity* in Fabbri’s 2024 essay, following an earlier manuscript by Fabbri and Oliveira Jr. (2017). The earlier manuscript assumes uniform resource allocation across component-wealth values; the later essay proposes a much broader natural and cosmological principle. The useful question is which allocation statement can be defined and tested independently of the abundance distribution it seeks to explain.

The ecological precedent is substantial. Sheldon, Prakash and Sutcliffe (1972) studied ocean particle size distributions; Gaedke (1993) explicitly related equal biomass in logarithmic classes to normalized biomass spectra; Hatton et al. (2021) reconstructed the ocean spectrum from bacteria

to whales. Arranz et al. (2022) studied variation among hundreds of lake fish communities and its ecological correlates. Variation among individual systems is therefore already a research topic. Energetic equivalence (Damuth, 1981) is related, but its species-population sampling unit differs from community abundance per size interval. Species richness across sizes prevents automatic interchange of those statements.

Power laws arise under different generative mechanisms, and apparent straight lines in logarithmic plots do not establish a power-law distribution (Newman, 2005; Clauset, Shalizi and Newman, 2009). Rewriting a distribution as inverse resource cost does not identify a new mechanism. Our contribution is a critical reconstruction: we specify the measure and resource, derive their conditional consequences, give counterexamples to stronger interpretations, and assess what the repository’s existing data can support. We make no priority claim for the accounting identity or for ecological biomass equivalence.

2. Mathematical formulation

2.1 Objects, resources, and a reference measure

Let k be a positive size coordinate on a finite domain $D = [a, b]$, with $0 < a < b < \infty$. Choose a reference measure μ , with $0 < \mu(D) < \infty$, for the classes being compared. Let $n_\mu(k) = dN/d\mu$ denote object abundance and $\bar{q}(k) = E[q \mid k] > 0$ the conditional arithmetic mean of an additive resource per object. Then

$$\mathcal{O}_\mu(k) = \frac{dR}{d\mu} = \bar{q}(k)n_\mu(k). \quad (1)$$

For an observed class, the corresponding identity is simply $\sum_i q_i = N_{\text{class}}\bar{q}_{\text{class}}$. Resources may be a standing stock, such as biomass, or an accumulated event quantity over a declared observation period. The latter is not a stock resident in a size class, and neither is a flux through size classes. Additivity is needed for (1); a dynamical conservation law is not.

Define **orthopolity relative to μ** , denoted $\mathcal{O}_\mu$, by

$$\mathcal{O}_\mu(k) = C > 0 \quad \text{for } \mu\text{-almost every } k \in D. \quad (2)$$

This is an assumption about allocation. It gives $n_\mu = C/\bar{q}$. Equivalently, if an object is sampled uniformly from the population, its size density with respect to μ is

$$p_N(k) = \frac{1}{Z\bar{q}(k)}, \quad Z = \int_D \frac{d\mu(k)}{\bar{q}(k)} < \infty. \quad (3)$$

Sampling instead in proportion to the resource held gives $p_R(k) = \bar{q}(k)p_N(k)/E_N[q] = 1/\mu(D)$. Thus uniform resource-weighted size and inverse-cost object abundance are equivalent statements, under the same reference measure. This is an accounting equivalence, not a derivation of the allocation assumption. The count, resource total, and constants also satisfy $N = CZ$ and $R = C\mu(D)$. For a finite observed population, the identity is exact for class totals; a smooth density is a population model or approximation.

2.2 The measure changes the exponent

Write $u = \ln(k/k_0)$ for a fixed reference size k_0 . For $\bar{q}(k) = q_0(k/k_0)^d$, with $d > 0$, two different hypotheses give:

Equal resource with respect to	Resource spectrum	Count density with respect to dk
Linear size, $d\mu = dk$	$\bar{q}(k)dN/dk = C$	$dN/dk \propto k^{-d}$
Logarithmic size, $d\mu = du = dk/k$	$\bar{q}(k)dN/du = C$	$dN/dk \propto k^{-(d+1)}$

The second row follows because $dN/du = k dN/dk$. For biomass with $q = k = m$, the predictions are consequently density exponents 1 and 2, respectively. The 2017 manuscript's explicit uniform density over a linear wealth interval is not interchangeable with the logarithmic convention used in the empirical analyses below. Adopting logarithmic classes is a substantive specification, motivated by size-spectrum practice.

Logarithmic allocation is unchanged by changing size units or logarithm base, apart from a constant density multiplier. It also survives $z = (k/k_0)^c$ for constant $c > 0$, since $d \ln z = c d \ln k$. It is not invariant under arbitrary nonlinear coordinates. In general, $dR/dv = (dR/du)|du/dv|$. Choosing a transformation after observing R can therefore manufacture flatness; the coordinate and measure must be fixed independently.

Under $O_{\log}$ and the power-cost assumption, the slope of abundance per log interval is $-d$, the ordinary density exponent is $d + 1$, and the survival function on $[a, b]$ is

$$P(K \geq k) = \frac{k^{-d} - b^{-d}}{a^{-d} - b^{-d}}. \quad (4)$$

The familiar survival exponent d and rank-size exponent $1/d$ are approximations away from the upper boundary, not exact finite-domain identities. For an unbounded domain with $d > 0$, count can be finite while logarithmically uniform resource has infinite total. Finite boundaries are essential to a finite-budget interpretation.

2.3 Allocation postulates and limits of the accounting

Resource symmetry as a candidate law. The proposed rationale is that, absent restrictions favouring particular resource costs, allocation should not distinguish those costs. A natural law can adopt this as a physical postulate without a deeper mechanistic derivation. One precise sufficient formulation uses K declared classes with fixed positive per-object costs q_i , allocations $R_i = q_i N_i$, and fixed total $\sum_i R_i = R > 0$. If the probability law of $(R_1, \dots, R_K)$ is invariant under exchanging allocations across the fixed classes, then symmetry and the sum give

$$E[R_i] = R/K, \quad E[N_i] = R/(Kq_i).$$

This postulate selects resource allocations as the symmetric quantities; symmetry of counts would give a different prediction. It is stronger than conservation and is not implied by an unspecified absence of constraints. Indivisible objects may require an approximation where exact exchangeability is infeasible. The class measure also matters: with cost itself as coordinate, equal resource per dq predicts $dE[N]/dq \propto q^{-1}$, whereas equal resource per $d \ln q$ predicts $dE[N]/dq \propto q^{-2}$. Transformation-based treatments of indifference likewise require a specified symmetry (Jaynes, 1968); an inferential rule does not itself establish a physical law.

Equal expectation does not establish equal snapshots or relaxation toward equality. In a divisible-resource idealization, assigning all resource to one uniformly chosen class has the required symmetry and unequal allocation in every realization. Time-average equality needs an additional condition connecting time averages to expectations or its own empirical postulate. A test must therefore specify what "tends to allocate" means and identify eligible systems through conditions

independent of their observed allocation. A deeper mechanism is optional for an empirical law; the scope and predictive evidence are not.

Conservation does not imply equal allocation. On any bounded logarithmic domain, $\mathcal{O}(u) = A \exp(\beta u)$ can be normalized to the same resource total for every β . Conservation fixes the integral and leaves its shape undetermined. Likewise, maximum entropy over finitely many equally weighted object classes, constrained by a mean cost, yields $p_j \propto \exp(-\lambda q_j)$. Different reference measures or constraints give different distributions (Jaynes, 1957; Visser, 2013).

Equipartition does not imply a power law without a cost law. If $\bar{q}(k) = q_0 \exp(k/k_0)$, then $O_{\log}$ gives $dN/du \propto \exp(-k/k_0)$ and $dN/dk \propto \exp(-k/k_0)/k$. Neither is a power law. Geometric dimension supplies d only in a specified geometric model such as similar objects with volume proportional to length cubed. It does not establish the number of independent resource inputs in an arbitrary system.

Independent inputs do not generally add tail exponents. For example, two independent unit-scale Pareto variables with densities hx^{-h-1} , $x \geq 1$, $h > 0$, have product density $h^2 z^{-h-1} \ln z$, $z \geq 1$. Integration over the possible factorizations introduces a logarithmic factor. This limits the independent-input argument in the 2017 essay.

The resource must not be fitted to force the conclusion. For any positive abundance, defining $\bar{q} = C/n_\mu$ makes (2) hold identically. A test must instead specify the resource through measurements, definitions, or an independently justified cost model. An unrestricted residual multiplier called “friction” is equally uninformative unless its form or parameters are predicted independently.

2.4 What an ensemble statement would mean

Suppose a system has a power-shaped resource spectrum $\mathcal{O}_i(u) = A_i \exp(\beta_i u)$. A proposed ensemble condition $E[\beta_i] = 0$ concerns mean slopes. It does not imply $E[\mathcal{O}_i(u)]$ is constant. With equal $A_i = A$ and $\beta_i \sim N(0, \tau^2)$,

$$E[\mathcal{O}_i(u)] = A \exp(\tau^2 u^2 / 2), \quad (5)$$

which is curved whenever $\tau > 0$. Equal-total normalization changes this formula but does not make mean-zero slopes sufficient: averaging normalized spectra proportional to $\exp(cu)$ and $\exp(-cu)$ on a symmetric domain produces a nonconstant hyperbolic cosine. A median slope is a different estimand again. A claim about average resource occupancy requires spectra, a common domain, and an explicit weighting of systems; slopes alone do not identify it.

More generally, let $H(u) = \sum_{i=1}^n w_i A_i e^{\beta_i u}$ on a common log-domain, with fixed positive weights and amplitudes. Then

$$H''(u) = \sum_i w_i A_i \beta_i^2 e^{\beta_i u} \geq 0.$$

If any slope is nonzero, the derivative is strictly positive, so the arithmetic mixture cannot be constant on an open interval. Exact flatness requires every contributing slope to be zero. With $p_i(u) = w_i A_i e^{\beta_i u} / H(u)$, its log-curvature is $d^2 \ln H / du^2 = \text{Var}_{p(u)}(\beta)$. Per-system total normalization preserves these conclusions. They are standard convexity consequences for exact power spectra; arbitrary curved profiles can cancel to a flat mean. They therefore neither exclude a general resource-symmetry law nor convert noisy fitted slopes into a profile test. The scientific assessment gives a quantitative bound for approximate flatness and its weighting restrictions.

Mean and variance also do not require a Gaussian distribution. Gaussianity is an additional maximum-entropy modelling choice, and selecting it by AIC is not a goodness-of-fit test. If

$\beta_i \sim N(\mu_\beta, \tau^2)$ and reported slope error is independent $N(0, \sigma_i^2)$, the predicted probability that the *reported slope* lies in $[-c_i, c_i]$ is

$$\Phi_N\left(\frac{c_i - \mu_\beta}{\sqrt{\tau^2 + \sigma_i^2}}\right) - \Phi_N\left(\frac{-c_i - \mu_\beta}{\sqrt{\tau^2 + \sigma_i^2}}\right). \quad (6)$$

The expression $2\Phi_N(c_i/\tau) - 1$ applies to zero-centred *latent* slopes. Comparing it to observed estimates without their errors is a different calculation. Fitting τ on the same slopes makes either comparison an internal diagnostic, not independent validation.

There is a valid simultaneous-resource restriction for an individual system: $\mathcal{O}_1/\mathcal{O}_2 = \bar{q}_1/\bar{q}_2$. Both spectra can be exactly flat only if their mean resource ratio is size-independent. Across systems, however, $\beta_{1i} - \beta_{2i} = d_{1i} - d_{2i}$ for power cost laws. Equal variances and correlation one follow only if that difference is constant and slope variance is nonzero, whether or not equipartition holds. They are not a general test of ensemble equipartition.

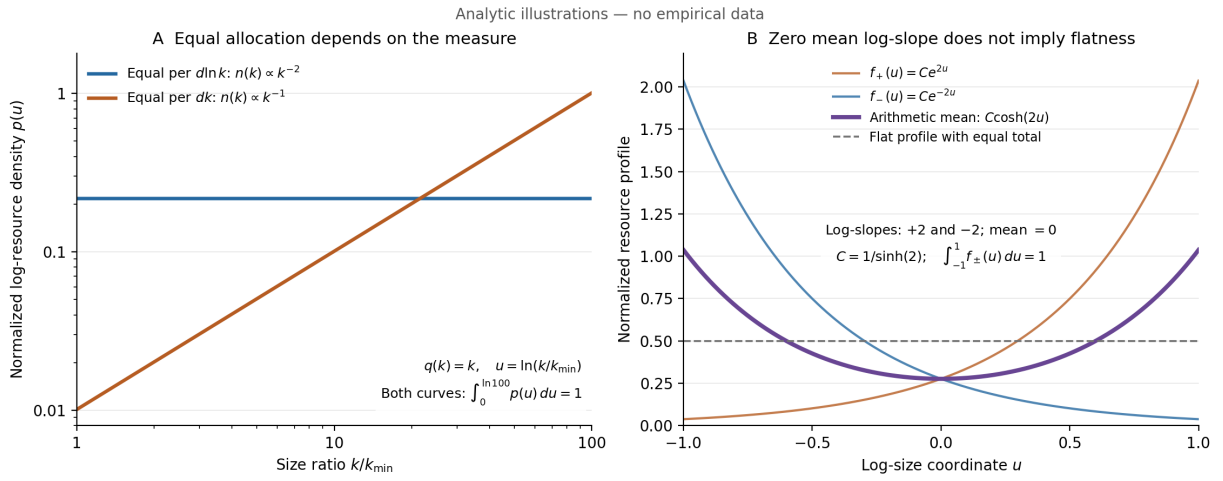


Figure 1. Analytic illustrations, not empirical data. (A) With resource equal to size, linear and logarithmic resource equipartition produce different count exponents. Both curves show unit-total resource density in the logarithmic coordinate. (B) Two unit-total resource profiles on a common domain have opposite log-slopes and therefore zero mean log-slope, but their arithmetic mean is not flat. Generated by `run_theory.py`; a vector version is available.

3. Data and methods

3.1 Sources and scope

Data source	Analysed material	Resource and size coordinate
NOAA GOES XRS flare reports, 2022–2024	10,501 catalogue events; 1,168 paired events in the primary evaluation domain	End fluence in J/m ² ; peak irradiance in W/m ²
USGS ComCat, 2010–2024	6,639 events after retaining moment-magnitude types and rounding to 0.1 at threshold 5.5	Magnitude-derived energy proxy as both resource and coordinate
Hatton et al. (2021)	23 one-decade biomass bins, upper 200 m and full water column	Body mass for both
GLOSSAQUA (Ersoy et al., 2025)	1,300 normalized biomass-spectrum estimates from 16 study identifiers	Body mass for both

These are four data sources; the two ocean domains are related reconstructions. GLOSSAQUA contains published estimates rather than organism-level observations. Study identifiers do not guarantee independent authors, methods, sites, or measurement errors. The sources are convenience cases for evaluating a hypothesis, not a representative sample of natural systems. Input bytes are fixed by SHA-256 manifests; provenance and source notices accompany the data.

3.2 Analysis-plan provenance

The pilot configuration explicitly labels itself exploratory. The subsequent aquatic/ocean plan is retained as prereg_2026-09-09.json. In local history, commit 6df364f contains the plan and raw slope data, preceding analysis commit a0c9973. This records an order of commits. It does not establish an immutable public registration, independently verified blinding, or when values were first viewed. Statements of blinding in the historical file are author reports. The current corrections are retrospective, and the later ensemble, stratification, and variance analyses are exploratory. No claim of confirmatory preregistration is needed for the descriptive and mathematical results here.

3.3 Accounting and uncertainty

For bins B_j of logarithmic width w_j , compute $\hat{\mathcal{O}}_j = \sum_{i \in B_j} q_i / w_j$ and $\hat{C} = \sum_j \sum_{i \in B_j} q_i / \sum_j w_j$. The normalized profile $\phi_j = \hat{\mathcal{O}}_j / \hat{C}$ has width-weighted mean one by construction. This is not evidence of flatness. Empty bins remain in the profile, the final boundary is included, and missing resources require an explicit exclusion policy.

A factor- F endpoint drift across log-domain width L corresponds, for a power-shaped spectrum, to slope tolerance $c = \ln(F)/L$. This differs from requiring every $\phi_j \in [1/F, F]$: the latter permits a maximum-to-minimum ratio of F^2 . Both slope and profile deviations matter. A slope equivalence interval combined with a point-estimate profile screen is not full simultaneous equivalence; that would require uncertainty for the whole profile. Failure to establish equivalence is not, by itself, evidence of a nonzero departure.

For flares, the arithmetic mean resource exponent is fitted by Gamma quasi-likelihood on 2022 data, with bounded count-density fitting on 2023–2024 over $[10^{-5}, 10^{-3}]$ W/m². Log-resource least squares would target a different conditional quantity unless an appropriate retransformation were justified. Pilot intervals use 600 calendar-month block resamples for flares and calendar-year

blocks for earthquakes. These do not capture all dependence or selection bias. The temporal split separates estimation samples but does not make the exploratory domain choice a prospective prediction.

Power-law diagnostics use maximum likelihood and a refitted parametric-bootstrap KS statistic on declared support, following the approach of Clauset et al. (2009). The rounded earthquake magnitudes are tested on their integer grid: for $K = (M - M_0)/\Delta$, $P(K = k) = (1 - r)r^k$ with $r = 10^{-b\Delta}$. Continuous-reference KS calibration is unsuitable for these ties. Parametric-bootstrap probabilities remain conditional on an independent-event working model; catalogue clustering is a limitation.

3.4 Aquatic estimates and reconstruction uncertainty

The primary GLOSSAQUA subset requires body mass on the size axis, finite slopes and positive ordered size limits, and the normalized biomass-spectrum convention. For biomass per unit linear mass $B(m) \propto m^t$, equal biomass per log mass requires $t = -1$. We examine $\beta = t + 1$ and each estimate's reported size span. The historical plan uses $F = 1.25$ and $F = 2$, a 2,000-replicate study-block bootstrap for the pooled median, and a joint criterion involving its interval and the fraction of slopes inside their tolerances. That fraction ignores individual slope uncertainty and must not be interpreted as formal individual equivalence. Slope summaries cannot detect within-spectrum curvature. A post-audit sensitivity explicitly excludes StudyID_07 from range-dependent assessment because its reported bounds are physically implausible; no replacement bounds are imputed (§4.4).

For the ocean reconstruction, 20,000 simulated profiles propagate reported multiplicative uncertainty factors using lognormal draws centred on the published log estimates, with log-scale standard deviation $\ln(f)/1.96$. The published bounds match estimate divided and multiplied by f . Independent errors across groups and bins are a working assumption, not a property established by those bounds. The reconstruction is not a sample of independent organisms, and its plateau was selected after inspecting the upper-ocean estimates.

Exploratory random-effects and latent-distribution fits use reported errors where available. Intervals converted to standard errors assume their advertised coverage and approximate normality. Size-spectrum estimators can have biased estimates and miscalibrated intervals (Edwards et al., 2017, 2020). Repeated spectra within studies and estimated error variances further limit inference; independence-based pooled intervals are not primary evidence of a population mean.

4. Results

4.1 Earthquakes contradict the specified energy-proxy allocation

Gutenberg–Richter scaling $N(\geq M) \propto 10^{-bM}$ and proxy $E \propto 10^{\gamma M}$ give resource per log energy proportional to $E^{1-b/\gamma}$. Flatness therefore requires $b = \gamma$. At threshold 5.5, the fitted $b = 0.998$ has year-block 95% interval $[0.973, 1.024]$, far below the chosen $\gamma = 1.5$. Thresholds 6.0 and 6.5 give 0.984 and 0.977. Using conversion exponent 1.44 also leaves a substantial discrepancy.

At threshold 5.5 the discrete KS statistic is approximately 0.0077, with a parametric-bootstrap probability about 0.26. This is non-rejection of the fitted geometric model, not proof that the model is exact. A power-like tail can thus coexist with a resource-proxy allocation that disagrees with $O_{\log}$. No discrete lognormal comparison was conducted. Magnitude-derived energy is not independently measured radiated energy; the finding is restricted to the specified proxy and catalogue, without declustering or a regional completeness model.

4.2 Flares disagree with the joint prediction on the selected domain

The fitted resource exponent is $d = 0.858$ with 95% interval $[0.697, 1.054]$. The implied density exponent is 1.858, compared with 2.239 $[2.085, 2.399]$ in the paired evaluation sample. The estimated gap is 0.382 $[0.125, 0.620]$. A rise-phase fluence sensitivity gives gap 0.411 $[0.208, 0.604]$. This alternate resource has no missing values in the domain, but does not prove the primary end-fluence analysis free of missingness bias.

For all 1,306 in-domain evaluation events, peak irradiance has fitted exponent 2.274 and KS statistic about 0.032; its bootstrap probability is about 0.02. The power law is therefore inadequate under this working test. Its likelihood cannot clearly distinguish it from the fitted lognormal. Exponent mismatch tests the conjunction of equipartition, power-shaped mean cost, transfer between years, and the count model; it cannot isolate equipartition from all those assumptions.

The gaps at four lower cutoffs are -0.480 , 0.191 , 0.382 , and 0.960 . This substantial sensitivity discourages treating one selected range as a universal scaling regime. Fluence is instrument-band irradiance integrated over an event, not total flare energy.

4.3 The ocean reconstruction is compatible with a broad, uneven plateau

The upper-ocean summary reproduces the published abundance slope at approximately -1.039 . Its maximum-to-minimum biomass ratio is 38.8 across all 23 bins, and 1.7 within the selected plateau of 16 bins. The corresponding full-water-column plateau ratio is 4.3. The 23 one-decade bins span 23 decades between outer edges; their centres are 22 decades apart. Similarly, the plateau covers 16 decades between edges and 15 between centres.

Under the stated uncertainty propagation, only 2 of 23 upper-ocean bin intervals lie wholly outside the factor-1.25 band; the corresponding full-column count is 3. These are pointwise intervals without multiplicity adjustment. The remaining intervals can include both near-flat and materially non-flat values, so their overlap with the band establishes neither equality nor absence of departures. Correlated reconstruction errors and post hoc range selection further limit the assessment. These results re-express Hatton et al.’s reconstruction; they do not independently replicate it.

The full upper-ocean range has a fitted biomass log-slope of -0.0392 and a propagated 90% interval approximately $[-0.058, -0.026]$, outside the factor-1.25 slope tolerance of 0.00421. Thus the assumed uncertainty model indicates a systematic gradient even though most individual bin intervals overlap the pointwise band. Pointwise overlap does not make the whole-profile question uninformative; its interpretation remains conditional on the reconstruction error model.

4.4 Aquatic slopes are close in location, without established equivalence

Among 1,300 normalized biomass spectra from 16 study identifiers, the pooled median slope is -1.015 . The study-block 95% interval for its departure from -1 is $[-0.100, 0.010]$. The median slope tolerance is 0.0317 at $F = 1.25$ and 0.0984 at $F = 2$. This interval is not wholly within either band. Under the historical joint criteria, the result is **not supported**.

The reported-slope fractions inside their individual drift tolerances are 0.07846 ($F = 1.25$) and 0.24231 ($F = 2$), counting boundary cases inclusively. These are descriptive compatibility fractions. Their complements are not estimates of the fraction of true spectra violating equipartition, because measurement error and within-spectrum shape are not accounted for.

These historical fractions also depend on a metadata defect: all 377 StudyID_07 records report bounds of 2×10^{-8} to 2×10^{27} pg C, a 35-decade range with a physically implausible upper body

mass of 2×10^{12} kg C. Excluding those records from this range-dependent calculation leaves 923 slopes from 15 studies. Their point-slope compatibility fractions are 9.32% at $F = 1.25$ and 28.93% at $F = 2$. The source-derived replacement bounds are unknown; the original slopes remain in the location summary. The remaining records also require a methods and units audit. Consequently neither version of the pass fraction should be treated as a validated estimate of ecological prevalence.

Two studies supply 1,016 of the 1,300 estimates, approximately 78%. A central estimate near -1 is worth investigating, but it neither establishes a mean of -1 in a defined population nor establishes equal average biomass occupancy. The stronger ensemble interpretation fails on both statistical and mathematical grounds (§2.4).

4.5 Exploratory diagnostics identify further limits

Latent-shape fits and random-effects summaries describe appreciable variation in the error-reporting subset. Their fitted dispersion includes ecological differences, study methods, and potentially unmodelled dependence. An I^2 estimate is not a measured fraction of variance caused by ecosystem differences. AIC rankings among Gaussian, Laplace, and Student families do not establish adequacy or identify ecological classes. Error-aware, matched-sample pass fractions are internal model checks, not independent predictions of individual failure.

The cross-study comparison of 639 lake-fish estimates with 377 records associated with Gaedke’s Lake Constance study does not partition spatial and temporal variance. Different taxa, sampling designs, fitting methods, and measurement uncertainties prevent treating their variance ratio as a bound on one population. Reproducing Arranz et al.’s reported summary statistics from the same underlying observations checks extraction, not independent replication of a dispersion parameter.

Exponent metadata also need source verification. Counts in equal-log bins and abundance divided by linear bin width have slopes differing by one under power scaling. The Perkins et al. (2018) count-bin method illustrates why a database category alone may not identify the estimand. Choosing a mapping because it makes a slope closer to -1 or -2 would bias the test. Secondary conventions remain sensitivity analyses; a complete study-level methods audit is needed before pooling them. Checks of two dominant primary studies do not validate every record in the primary subset.

5. Discussion

The conditional identity is useful because it forces a researcher to name the resource, sampling unit, coordinate, and reference measure before interpreting a slope. It also separates a distributional question from an allocation question: a power-law model can fit without the specified resource being equally allocated. Conversely, equal allocation can hold with a non-power cost law and yield a different abundance shape.

Neither this identity nor the current case studies establishes a universal tendency. The original linear-allocation assumption and the ecological logarithmic-allocation assumption must be treated separately. The physical examples contradict selected proxy versions, while the ecological summaries contain suggestive locations and substantial uncertainty. No result here explains why a dynamical system should approach equal allocation, proves an equilibrium or time arrow, or quantifies the work needed to change social inequality. No spacetime model or cosmological prediction has been supplied.

A narrower empirical claim remains possible. It would specify a population of systems, one independently justified resource and measure, a sampling frame, a common or explicitly varying

size domain, and an equivalence tolerance with scientific meaning. Raw resource totals would permit testing profile shape as well as slope. A design with multiple sites and repeat visits could separate sources of variation using a model appropriate to its sampling structure. A new dataset could test a frozen conditional prediction, including its uncertainty; numerical proximity to a dispersion fitted on the original sample is not sufficient.

The principal limitation is the absence of independently specified conditions identifying the proposed neutral regime and of data/design sufficient to test its generalization. An empirically supported allocation law need not explain its symmetry through a deeper mechanism. Its predictions must nevertheless specify the resource, class measure, and statistical target. An experiment that weakens an independently measured source of bias could test approach to the neutral prediction, if that convergence is part of the declared hypothesis. Departures cannot simply be relabelled as previously unspecified restrictions.

Prior ecological work already supplies models and environmental explanations (Cuesta, Delius and Law, 2018; Arranz et al., 2022). Further orthopolity research needs a demonstrable contribution relative to those accounts, such as an independently validated conditional prediction or a calibrated diagnostic comparison. A resource-symmetry law could coexist with mechanisms explaining when it applies and how it is approached.

6. Conclusion

Orthopolity can be stated as a mathematically coherent, falsifiable hypothesis once its resource, measure, and domain are fixed. Its inverse-cost consequence is an accounting identity shared with established size-spectrum theory. Resource symmetry provides a coherent postulate for equal expected allocations; a deeper mechanism is not required for it to become an empirically established law. The stronger universal, cosmological, and ensemble-confirmation claims are not justified by the available arguments or evidence. A scientific document is warranted as a critical synthesis with reproducible exploratory tests and explicit counterexamples. A paper announcing a new natural law is not warranted.

Data and code availability

Code, frozen inputs, configurations, and generated results are in this repository. The make all target verifies input checksums, runs tests, and regenerates the empirical analyses; see the README for setup. Pilot estimates originate in results.json; distribution diagnostics in gof.json; aquatic median, compatibility fractions, and ocean intervals in independent.json. Remaining JSON files contain explicitly exploratory diagnostics. Source-specific conditions are recorded in data/NOTICE.md; there is no blanket licence for all inputs. The supplied 2017 manuscript was inspected privately and is not redistributed.

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